

Information-Topological Theory: Minimal Foundational Formulation

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Abstract

We present the minimal foundational formulation of Information-Topological Theory (ITT) as a timeless canonical gauge framework on a three-dimensional configuration space. The formulation separates standard mathematical results from ITT-specific postulates, completed intermediate derivations, negative mathematical results, and future dynamical predictions.

The standard sector contains the $SU(2)$ gauge structure on S^3 , the canonical ITT constraint, its formal Chern–Simons state, the unit BPST topological sector, the associated Atiyah–Patodi–Singer spectral-flow construction, and ordinary integer Chern–Simons level quantization. ITT further postulates a closed-wave ontology and a unit scale-ratio self-similarity condition $\ell/R = 1$; elementary chord geometry then yields the sixfold regular cyclic closure $N = 6$. The derived cyclic order is realized by the specified free diagonal $\mathbb{Z}_6$ action on S^3 , giving the quotient $L(6, 1)$ and its convention-fixed rational Chern–Simons invariant. The level- k differential Chern–Simons evaluation of this sector gives the phase $\exp(2\pi i k \text{cs}(\rho_1)) = e^{-2\pi i k/6}$. A further analysis shows that the presently specified data do not canonically identify this phase with a stabilizer action on a hypothetical three-dimensional prequantum line fiber. ITT therefore formulates level selection as an explicit phase-triviality postulate rather than as an unproved equivariant-line-bundle theorem.

Under that theory-specific selection postulate, $Z_k(L(6, 1), \rho_1) = 1$, one obtains $k \in 6\mathbb{Z}$ and hence $k_{\min} = 6$. No Standard-Model parameter fitting or generation-count argument is used in the foundational level selection.

1 Scope and Logical Status

The Information-Topological Theory (ITT) is formulated as a timeless canonical gauge theory on a three-dimensional spatial manifold. The fundamental configuration space is the gauge-orbit space

$$\mathcal{C}_{\text{ITT}} = \mathcal{A}(S^3)/\mathcal{G}(S^3), \quad (1)$$

where $\mathcal{A}(S^3)$ denotes the space of smooth $SU(2)$ connections on S^3 and $\mathcal{G}(S^3)$ the corresponding group of gauge transformations.

No preferred external physical time is postulated at the foundational level. A parameter used to label a family of configurations is therefore not identified with physical time.

The theory separates four logical levels:

- (i) **Standard mathematics:** gauge theory on S^3 , BPST topology, APS spectral flow, ordinary Chern–Simons level quantization, lens-space topology, and rational Chern–Simons invariants.
- (ii) **ITT defining structure:** the timeless canonical constraint, the closed-wave physical ontology, the unit scale-ratio self-similarity condition, and the phase-triviality requirement used for level selection.

- (iii) **Completed intermediate derivations:** the sixfold regular cyclic closure $N = 6$, the differential Chern–Simons phase, and its induced cyclic character associated with the resulting order-six quotient realization.
- (iv) **Open or future ITT results:** dynamical derivation of $N_{\text{gen}} = 3$, precision phenomenology, and any stronger physical interpretation of the selected level.

The explicit three-dimensional equivariant-line interpretation is not assumed as an established theorem. Paper 1 records the mathematical obstruction to obtaining such an identification from the presently specified data.

2 Canonical Configuration Space and Conventions

2.1 Gauge Variables

The canonical variables are

$$A_i^a(x), \quad E_a^i(x), \quad (2)$$

with fundamental Poisson brackets

$$\{A_i^a(x), E_b^j(y)\} = \delta_i^j \delta_b^a \delta^{(3)}(x - y), \quad (3)$$

and

$$\{A_i^a(x), A_j^b(y)\} = 0, \quad \{E_a^i(x), E_b^j(y)\} = 0. \quad (4)$$

The momentum E_a^i is taken to be a densitized spatial vector of weight $+1$. The $SU(2)$ generators are normalized by

$$T_a = \frac{\sigma_a}{2i}, \quad \text{tr}(T_a T_b) = -\frac{1}{2} \delta_{ab}, \quad [T_a, T_b] = \epsilon_{abc} T_c. \quad (5)$$

The curvature is

$$F = dA + A \wedge A, \quad (6)$$

and the magnetic density is

$$B_a^i = \frac{1}{2} \epsilon^{ijk} F_{jk,a}, \quad (7)$$

where ϵ^{ijk} denotes the spatial Levi-Civita density.

3 Chern–Simons Functional and Fundamental ITT Constraint

Define the unnormalized Chern–Simons functional

$$S_{\text{CS}}^{(0)}[A] = \int_{S^3} \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right). \quad (8)$$

With the trace convention above, its functional derivative is

$$\frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a(x)} = -B_a^i(x). \quad (9)$$

The fundamental ITT constraint is

$$\boxed{C_a^i \equiv E_a^i + \frac{k}{4\pi} B_a^i = E_a^i - \frac{k}{4\pi} \frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a} \approx 0.} \quad (10)$$

The corresponding first-order action is

$$S_{\text{ITT}} = \int d\tau \int_{S^3} d^3x \left[E_a^i \frac{dA_i^a}{d\tau} - N_i^a C_a^i - A_0^a \mathcal{G}_a \right]. \quad (11)$$

Here τ is an auxiliary parameter labeling a family of configurations; it is not assumed to be physical time.

4 Constraint Algebra

Proposition 4.1 (Commuting ITT constraints). *On the closed manifold S^3 , the constraints*

$$\mathcal{C}_a^i = E_a^i - \frac{k}{4\pi} \frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a} \quad (12)$$

satisfy

$$\{\mathcal{C}_a^i(x), \mathcal{C}_b^j(y)\} = 0. \quad (13)$$

Proof. The term quadratic in functional derivatives vanishes because $\{A, A\} = 0$, while $\{E, E\} = 0$. The two mixed terms are opposite second functional derivatives of the same smooth functional $S_{\text{CS}}^{(0)}$. Since S^3 is closed, no boundary contribution remains, and equality of mixed functional derivatives gives exact cancellation. $\square$

The non-Abelian Bianchi identity gives

$$D_i B_a^i \equiv 0. \quad (14)$$

Therefore

$$D_i \mathcal{C}_a^i = D_i E_a^i \equiv \mathcal{G}_a. \quad (15)$$

Hence

$$\mathcal{G}_a = D_i \mathcal{C}_a^i. \quad (16)$$

Remark 4.2. *At the classical level the Gauss constraint is a differential consequence of the ITT constraint. The full set is naturally described as a first-class, differentially related constraint system. At the quantum level the corresponding identity requires a fixed gauge-covariant operator ordering.*

For the scalar constraint we use the densitized convention

$$\mathcal{H}_{\text{ITT}} = \frac{1}{2\sqrt{g}} g_{ij} \mathcal{C}_a^i \mathcal{C}^{ja}. \quad (17)$$

This definition fixes the density weight explicitly.

5 Formal Chern–Simons State

Canonical quantization is represented formally[3] by

$$\hat{E}_a^i = -i \frac{\delta}{\delta A_i^a}. \quad (18)$$

Define

$$\Psi_k[A] = \mathcal{N} \exp \left[i \frac{k}{4\pi} S_{\text{CS}}^{(0)}[A] \right]. \quad (19)$$

Then

$$\hat{E}_a^i \Psi_k[A] = -\frac{k}{4\pi} B_a^i \Psi_k[A], \quad (20)$$

so that formally

$$\hat{\mathcal{C}}_a^i \Psi_k[A] = 0. \quad (21)$$

Consequently, for the positive quadratic operator corresponding to the classical constraint,

$$\hat{\mathcal{H}}_{\text{ITT}} \Psi_k[A] = 0 \quad (22)$$

formally, subject to a specified regularization and operator ordering.

This is a formal exact solution of the ITT constraint system. A complete Hilbert-space construction, functional measure, self-adjoint realization, and ultraviolet regularization are separate analytical problems and are not claimed to be solved here.

6 Unit Topological Sector: BPST and APS

Consider the standard $SU(2)$ BPST instanton of scale ρ on flat $\mathbb{R}^4$, restricted to the four-ball D_R^4 [1]. Define

$$c = \frac{R^2}{R^2 + \rho^2}. \quad (23)$$

With the normalization above,

$$Q(R) = -\frac{1}{8\pi^2} \int_{D_R^4} \text{tr}(F \wedge F) = 3c^2 - 2c^3 = \frac{R^4(R^2 + 3\rho^2)}{(R^2 + \rho^2)^3}. \quad (24)$$

On the boundary,

$$A|_{S_R^3} = c g^{-1} dg, \quad (25)$$

and the Chern–Simons transgression gives, with the same orientation convention,

$$\frac{1}{2\pi} S_{\text{CS}}(A|_{S_R^3}) = 3c^2 - 2c^3. \quad (26)$$

Thus

$$\lim_{R/\rho \rightarrow \infty} Q(R) = 1. \quad (27)$$

Introduce the auxiliary cylinder

$$Y = [0, 1] \times S^3 \quad (28)$$

with product metric and the family

$$A(c) = c g^{-1} dg. \quad (29)$$

The parameter c labels a path in configuration space and is not physical time. For the unit-winding interpolation, the standard APS spectral-flow correspondence identifies the net spectral flow with the unit topological index[2]:

$$\boxed{\text{SF}(\mathcal{D}_{A(c)}) = \text{ind}(\mathcal{D}_Y^{\text{APS}}) = 1.} \quad (30)$$

The statement above concerns the net spectral flow. It does not assert, without a separate spectral calculation, that exactly one individual eigenvalue crosses zero.

7 Standard Chern–Simons Level Quantization

For a large gauge transformation

$$w \in \pi_3(SU(2)) \cong \mathbb{Z}, \quad (31)$$

the normalized Chern–Simons amplitude is gauge invariant only when the level satisfies

$$\boxed{k \in \mathbb{Z}.} \quad (32)$$

This is the standard Chern–Simons quantization condition[3]. It does not select $k = 6$.

8 Closed-Wave Ontology and Sixfold Structural Closure

The order of the cyclic quotient used below is not taken here as an unexplained numerical input. ITT introduces two explicit physical postulates and separates them from the geometry that follows.

Postulate 8.1 (ITT Closed-Wave Physical Ontology). *A stable elementary physical excitation is represented by a globally phase-coherent closed field mode. For a closed orbit of length L , the field satisfies*

$$\boxed{\Psi(s + L) = \Psi(s)}. \quad (33)$$

For a phase mode $\Psi(s) = Ae^{iqs}$, closure implies

$$qL = 2\pi n, \quad n \in \mathbb{Z}. \quad (34)$$

ITT takes the minimal non-trivial closed excitation to have fundamental winding

$$\boxed{n_{\min} = 1}. \quad (35)$$

Remark 8.2. *The integer-winding consequence is standard mathematics once periodic single-valuedness is imposed. The specifically physical statement that stable elementary excitations are to be modeled by such globally closed modes is the ITT postulate.*

Postulate 8.3 (ITT Unit Scale-Ratio Self-Similarity). *For the elementary regular cyclic discretization of the fundamental closed mode, let R denote the curvature radius of the circular phase orbit and let ℓ denote the straight chord joining adjacent phase points. ITT imposes the scale-free self-consistency condition*

$$\boxed{\eta \equiv \frac{\ell}{R} = 1}. \quad (36)$$

This is a defining ITT geometric postulate, not a theorem of standard quantum theory or differential geometry.

Theorem 8.4 (Sixfold structural closure). *For a regular cyclic closure with integer $N \geq 3$, the ITT unit scale-ratio postulate uniquely selects*

$$\boxed{N = 6}. \quad (37)$$

Proof. For a regular N -fold subdivision of a circle of radius R , adjacent phase points subtend the central angle $2\pi/N$, so the chord length is

$$\ell_N = 2R \sin\left(\frac{\pi}{N}\right). \quad (38)$$

Imposing $\ell_N = R$ gives

$$2 \sin\left(\frac{\pi}{N}\right) = 1. \quad (39)$$

For $N \geq 3$, one has $0 < \pi/N \leq \pi/3$. In this interval the equation $\sin x = 1/2$ has the unique solution $x = \pi/6$. Hence

$$\frac{\pi}{N} = \frac{\pi}{6}, \quad (40)$$

and therefore $N = 6$. □

Corollary 8.5 (Derived cyclic order). *The elementary regular closure selected by A-05 has cyclic order six and therefore carries the abstract cyclic symmetry*

$$\boxed{\mathbb{Z}_6 = \langle \zeta \mid \zeta^6 = 1 \rangle}. \quad (41)$$

Remark 8.6 (Logical boundary). *The theorem derives the integer order 6 from A-05; it does not by itself prove uniqueness of a free $\mathbb{Z}_6$ action on S^3 and does not independently derive the lens-space parameter $q = 1$. ITT realizes the derived cyclic order by the explicit free diagonal action on $S^3 \subset \mathbb{C}^2$ used in the next section. Thus the number six is no longer an independent numerical input, while the concrete quotient realization remains part of the ITT geometric construction.*

9 Lens-Space Sector

The derived cyclic order $N = 6$ is realized in the ITT quotient geometry by the free diagonal action

$$(z_1, z_2) \mapsto (e^{2\pi i/6} z_1, e^{2\pi i/6} z_2) \quad (42)$$

on $S^3 \subset \mathbb{C}^2$. Thus

$$\boxed{L(6, 1) = S^3 / \mathbb{Z}_6,} \quad (43)$$

and

$$\pi_1(L(6, 1)) \cong \mathbb{Z}_6. \quad (44)$$

Let ρ_1 be the fundamental non-trivial flat representation

$$\rho_1(\gamma) = \begin{pmatrix} e^{2\pi i/6} & 0 \\ 0 & e^{-2\pi i/6} \end{pmatrix}, \quad (45)$$

for a chosen generator γ of $\pi_1(L(6, 1))$.

Define the normalized Chern–Simons invariant by

$$\text{cs}(A) = \frac{1}{8\pi^2} \int_{L(6,1)} \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right) \pmod{\mathbb{Z}}. \quad (46)$$

With the orientation and trace convention fixed above, the fundamental flat sector has

$$\boxed{\text{cs}(\rho_1) = -\frac{1}{6} \pmod{\mathbb{Z}}.} \quad (47)$$

One way to obtain this value is through the (-6) -framed surgery description of $L(6, 1)$ [4, 5]. The inverse surgery matrix gives the rational boundary linking pairing $-1/6$. The fractional value is a three-dimensional secondary invariant.

Remark 9.1. *The rational value $-1/6$ is not an ordinary second Chern number on a closed four-manifold such as $L(6, 1) \times S^1$. Ordinary integral characteristic numbers remain integral.*

10 Differential Chern–Simons Phase

The level- k characteristic class is the integral class

$$kc_2 \in H^4(BSU(2); \mathbb{Z}), \quad (48)$$

with a differential refinement in the sense of Cheeger–Simons differential characters [7, 6]. For the flat connection (P_{ρ_1}, A_{ρ_1}) on the closed three-manifold $L(6, 1)$, its evaluation gives the exponentiated Chern–Simons invariant

$$\boxed{Z_k(L(6, 1), \rho_1) = \exp(2\pi i k \text{cs}(\rho_1)).} \quad (49)$$

Therefore

$$\boxed{Z_k(L(6, 1), \rho_1) = e^{-2\pi i k/6}.} \quad (50)$$

This is a derived consequence of the standard differential Chern–Simons construction together with the convention-fixed lens-space invariant; it is not an ITT postulate.

11 Cyclic Character Induced by the Fundamental Phase

Let

$$\Gamma = \mathbb{Z}_6 = \langle \zeta \mid \zeta^6 = 1 \rangle. \quad (51)$$

Since $k \in \mathbb{Z}$,

$$Z_k^6 = e^{-2\pi i k} = 1. \quad (52)$$

Hence the phase canonically determines, relative to the chosen generator ζ , a cyclic character

$$\boxed{\chi_k^{\text{CS}} : \Gamma \longrightarrow U(1), \quad \chi_k^{\text{CS}}(\zeta^n) = Z_k^n.} \quad (53)$$

In particular,

$$\boxed{\chi_k^{\text{CS}}(\zeta) = e^{-2\pi i k/6}.} \quad (54)$$

No statement is made that this character equals the stabilizer action of a separately constructed three-dimensional prequantum line; that identification is addressed as a negative result in the level-selection paper.

12 Revised ITT Level-Selection Postulate

The standard Chern–Simons construction supplies the phase Z_k , but does not by itself supply a canonical identification of that phase with a fiber character of a three-dimensional line object. ITT therefore makes the following explicit theory-specific choice.

Postulate 12.1 (ITT Fundamental-Sector Phase Triviality). *The fundamental quotient sector $L(6, 1)$ is admitted by ITT at level k only when its level- k Chern–Simons phase is trivial:*

$$\boxed{Z_k(L(6, 1), \rho_1) = 1.} \quad (55)$$

This postulate is stronger and more transparent than assuming an unconstructed equivariant line-bundle lift. It is an ITT-specific selection rule and is not claimed to follow from ordinary $SU(2)$ Chern–Simons theory.

13 Conditional Level Selection

Using the derived phase,

$$Z_k = e^{-2\pi i k/6}, \quad (56)$$

the phase-triviality postulate gives

$$e^{-2\pi i k/6} = 1. \quad (57)$$

Therefore

$$\boxed{k \in 6\mathbb{Z}.} \quad (58)$$

For positive integer level,

$$\boxed{k_{\min} = 6.} \quad (59)$$

The logical chain is therefore

$$\text{standard CS quantization } (k \in \mathbb{Z}) + \text{ITT phase-triviality postulate} \implies k_{\min} = 6. \quad (60)$$

14 Independent Soliton Sector

The number of generations is not used in the level-selection argument. ITT treats

$$N_{\text{gen}} = 3 \quad (61)$$

as an independent dynamical hypothesis to be tested by a future soliton analysis. The objective is to construct an ITT energy functional $E_{\text{ITT}}[n, A]$ for which the stable topological sectors satisfy

$$Q_H = 1, 2, 3, \quad (62)$$

while higher-charge configurations are unstable or energetically decomposable. Until such a stability theorem is obtained, $N_{\text{gen}} = 3$ remains an open ITT result and is not used to derive $k = 6$.

15 Foundational Logical Chain

The minimal ITT foundation is summarized by

$$\begin{aligned}
 & \mathcal{C}_{\text{ITT}} = \mathcal{A}(S^3)/\mathcal{G}(S^3) \\
 & \Downarrow \\
 & \boxed{\mathcal{C}_a^i = E_a^i + \frac{k}{4\pi} B_a^i, \quad \{\mathcal{C}, \mathcal{C}\} = 0} \\
 & \Downarrow \\
 & \boxed{D_i \mathcal{C}_a^i = \mathcal{G}_a} \\
 & \Downarrow \\
 & \boxed{\Psi_k[A] = \mathcal{N} e^{ik S_{\text{CS}}^{(0)}[A]/4\pi}} \\
 & \Downarrow \\
 & \boxed{Q_{\text{BPST}} \rightarrow 1, \quad \text{SF} = \text{ind}_{\text{APS}} = 1} \\
 & \Downarrow \\
 & \boxed{\Psi(s+L) = \Psi(s), \quad n_{\text{min}} = 1 \quad (\text{ITT closed-wave postulate})} \\
 & \Downarrow \\
 & \boxed{\ell/R = 1 \quad (\text{ITT self-similarity postulate})} \quad (63) \\
 & \Downarrow \\
 & \boxed{N = 6 \quad (\text{sixfold structural closure})} \\
 & \Downarrow \\
 & \boxed{L(6,1) = S^3/\mathbb{Z}_6, \quad \text{cs}(\rho_1) = -\frac{1}{6}} \\
 & \Downarrow \\
 & \boxed{Z_k = e^{2\pi i k \text{cs}(\rho_1)} = e^{-2\pi i k/6}} \\
 & \Downarrow \\
 & \boxed{Z_k = 1 \quad (\text{ITT phase-triviality postulate})} \\
 & \Downarrow \\
 & \boxed{k \in 6\mathbb{Z}} \\
 & \Downarrow \\
 & \boxed{k_{\text{min}} = 6.}
 \end{aligned}$$

The independent soliton problem begins after this foundational chain:

$$E_{\text{ITT}} \longrightarrow \text{stable Hopf sectors} \longrightarrow N_{\text{gen}} = 3. \quad (64)$$

16 Foundational Status and Open Problems

The present version deliberately does not claim completion of the full physical theory. The current status is:

Object	Status	Role
$SU(2)$ gauge geometry on S^3	Established	Foundation
BPST unit sector	Established	Topological input
APS spectral-flow framework	Established	Unit-index construction
$k \in \mathbb{Z}$	Established	Standard CS quantization
Closed-wave ontology, $n_{\min} = 1$	ITT Postulate	Physical ontology
$\ell/R = 1$	ITT Postulate	Geometric self-consistency
$N = 6$	Derived/Conditional	Consequence of $\ell/R = 1$
$L(6, 1) = S^3/\mathbb{Z}_6$	Constructed	Diagonal realization of derived order
$\text{cs}(\rho_1) = -1/6$	Derived	Convention-fixed CS input
$Z_k = e^{-2\pi i k/6}$	Derived	Differential-CS evaluation
χ_k^{CS}	Derived construction	Cyclic character
3D equivariant fiber identification	Obstructed	Paper 1 negative result
$Z_k = 1$	ITT Postulate	Level selection
$k_{\min} = 6$	Conditional	Consequence of postulate
$N_{\text{gen}} = 3$	Open	Soliton paper target
α , masses, $g - 2$, cosmology	Future	Prediction modules

The foundational theory therefore makes the restricted claim that the ITT postulate $\ell/R = 1$ selects the cyclic order $N = 6$ by elementary chord geometry; the diagonal $\mathbb{Z}_6$ action then provides the explicit quotient realization used by the theory. Standard differential Chern–Simons mathematics supplies the phase of that lens-space sector, while the phase-triviality condition is a separate ITT selection choice. No claim is made that either A-05 or $k = 6$ is a universal theorem of ordinary quantum or Chern–Simons theory.

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